CODEALPHA INTERNSHIP PROGRAM — BIOINFORMATICS

TASK 1: DNA / Protein Sequence Analysis & Homology Search

Project Identifier:	CodeAlpha_Task1_BLAST_Analysis
Target Molecule:	Human Preproinsulin (Homo sapiens)
Database Accession:	UniProt ID: P01308.1 / NCBI RefSeq: NP_000198.1
Analytical Tool:	NCBI Protein BLAST (blastp) — ClusteredNR Database

1. Introduction & Biological Context

Insulin is an essential peptide hormone synthesized by the beta cells of the pancreatic islets of Langerhans. It plays a pivotal role in maintaining glucose homeostasis by facilitating cellular glucose uptake in liver, adipose, and skeletal muscle tissues, while simultaneously inhibiting gluconeogenesis and glycogenolysis.

Human insulin is initially translated as a single-chain precursor termed preproinsulin, comprising 110 amino acids. Following signal peptide cleavage and post-translational processing, it yields active mature insulin consisting of an A-chain (21 amino acids) and a B-chain (30 amino acids) linked by interchain disulfide bonds. Performing sequence analysis and evolutionary homology searches on human preproinsulin allows bioinformaticians to identify highly conserved functional domains, key catalytic residues, and evolutionary conservation patterns across species.

2. Methodology & Computational Parameters

The complete protein sequence of human preproinsulin (Accession ID: P01308.1) was queried using the Basic Local Alignment Search Tool for Proteins (BLASTP) hosted by the National Center for Biotechnology Information (NCBI). BLAST algorithm calculates local sequence alignment metrics by identifying short matching words and extending them using scoring matrices.

Edit Search | Save Search | Search Summary

Search Parameters	
Program	blastp
Word size	5
Expect value	0.05
Hitlist size	100
Gapcosts	11,1
Matrix	BLOSUM62
Filter string	F
Genetic Code	1
Window Size	40
Threshold	0
Composition-based stats	2

Database	
Posted date	Jul 31, 2026 8:01 AM
Number of letters	171,756,544,853
Number of sequences	540,016,303
Entrez query	None

Karlin-Altschul statistics		
Lambda	0.323229	0.267
K	0.14055	0.041
H	0.452882	0.14
Alpha	0.7916	1.9
Alpha_v	4.96466	42.6028
Sigma		43.6362

Results Statistics	

Figure 1: BLASTP Algorithm Search Parameters

2.1 Configured BLAST Parameters

BLAST Program	blastp (protein-protein BLAST)
Target Database	ClusteredNR (540,016,303 sequences across all organisms)
Scoring Matrix	BLOSUM62 (Blocks Substitution Matrix)
Gap Penalty Costs	Existence: 11, Extension: 1
Expect Threshold (E-value)	0.05
Word Size	5
Compositional Adjustments	Conditional compositional score matrix adjustment

3. Results & Sequence Homology Analysis

The BLAST search executed against the ClusteredNR database yielded statistically significant alignments across primates, placental mammals, and rodents. The query sequence achieved maximum alignment scores with close evolutionary relatives, showing zero gaps and E-values approaching zero.

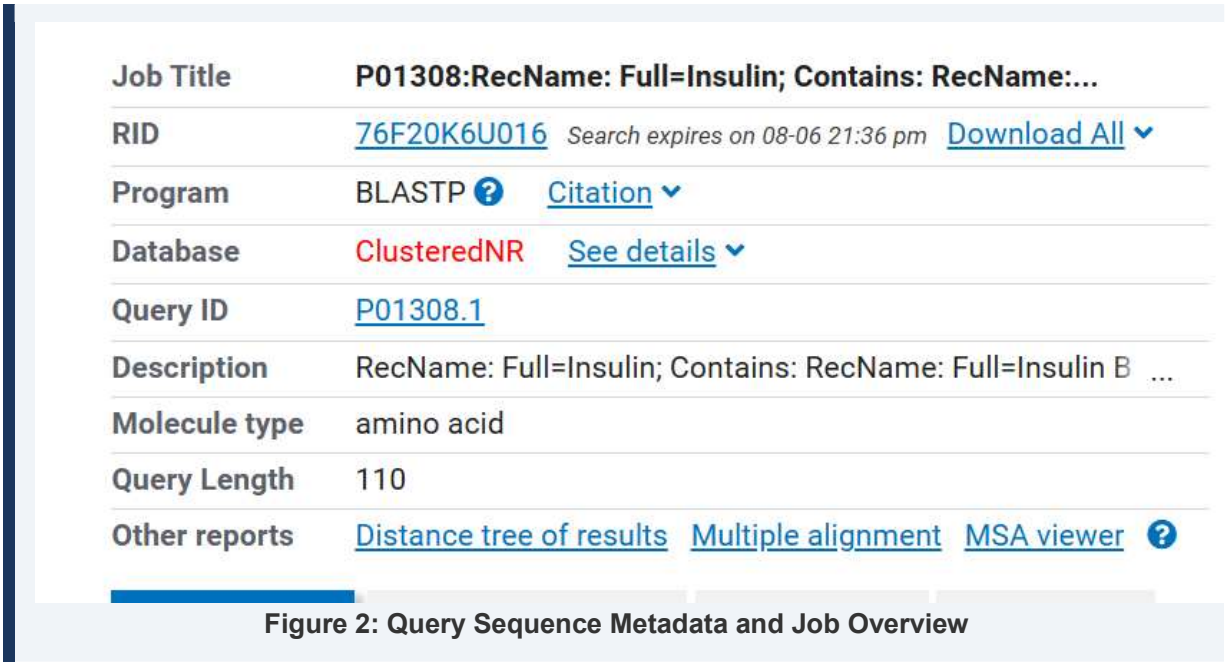


Figure 2: Query Sequence Metadata and Job Overview



Clusters producing significant alignments Download Select columns Show 100

select all *0 clusters selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Cluster Composition (Click the ⊞ to see the cluster contents)	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
⊞17 member(s), 11 organism(s)	primates	insulin preproprotein [Pan troglodytes]	221	221	100%	1e-72	98.18%	110	NP_001008996.1
⊞3 member(s), 3 organism(s)	primates	insulin [Pan paniscus]	224	224	100%	2e-72	99.09%	204	XP_034787832.1
⊞5 member(s), 5 organism(s)	primates	insulin isoform X1 [Macaca nemestrina]	221	221	100%	4e-72	98.18%	153	XP_011719619.2
⊞1 member(s), 1 organism(s)	green monkey	insulin isoform X2 [Chlorocebus sabaeus]	218	218	100%	5e-71	97.27%	144	XP_008002825.1
⊞1 member(s), 1 organism(s)	black snub-nosed monkey	insulin isoform X2 [Rhinopithecus bieti]	218	218	100%	1e-70	97.27%	147	XP_017743290.1
⊞1 member(s), 1 organism(s)	green monkey	insulin isoform X1 [Chlorocebus sabaeus]	218	218	100%	7e-70	97.27%	209	XP_008002752.1
⊞5 member(s), 5 organism(s)	primates	INS isoform 5 [Pongo abelii]	191	191	100%	1e-60	88.18%	98	PNJ16423.1
⊞2 member(s), 2 organism(s)	placentals	insulin [Tupaia tana]	184	184	100%	4e-58	90.00%	110	KAN7991649.1
⊞1 member(s), 1 organism(s)	European harvest mouse	insulin [Micromys minutus]	184	184	100%	5e-58	79.09%	110	XP_080768357.1
⊞1 member(s), 1 organism(s)	other sequences	ZZ-proinsulin [Cloning vector pET28a_ZZ_proinsulin]	183	183	78%	3e-56	100.00%	210	UFQ90408.1
⊞4 member(s), 4 organism(s)	bats	insulin [Rhinolophus affinis]	178	178	100%	2e-55	88.18%	110	KAN3685427.1
⊞1 member(s), 1 organism(s)	high G+C Gram-positive b…	hypothetical protein [Streptomyces chartreusis]	179	179	78%	2e-55	100.00%	143	MCZ4612290.1
⊞2 member(s), 1 organism(s)	other sequences	HPPI [synthetic construct]	175	175	78%	1e-54	98.84%	97	AGG11869.1
⊞member(s), organism(s)	Rhesus monkey	insulin pro- [Macaca mulatta]	174	174	78%	3e-54	98.84%	86	1006230A
⊞1 member(s), 1 organism(s)	reed vole	insulin [Alexandromys fortis]	175	175	100%	6e-54	85.45%	128	XP_049992438.1
⊞1 member(s), 1 organism(s)	Alston's brown mouse	hypothetical protein STEG23_014591 [Scotinomys teguina]	175	175	100%	9e-54	85.45%	145	KAL6031003.1
⊞1 member(s), 1 organism(s)	other sequences	SUMO-Lispro proinsulin fusion protein [synthetic construct]	176	176	78%	1e-53	97.67%	207	UIK24472.1
⊞3 member(s), 2 organism(s)	rabbits and hares	insulin precursor [Oryctolagus cuniculus]	173	173	100%	2e-53	86.36%	110	NP_001353985.1
⊞1 member(s), 1 organism(s)	coppery titi	Insulin [Plecturocebus cupreus]	175	175	100%	2e-53	87.27%	184	KAL0628984.1

⊞1 member(s), 1 organism(s)	reed vole	insulin [Alexandromys fortis]	175	175	100%	6e-54	85.45%	128	XP_049992438.1
⊞1 member(s), 1 organism(s)	Alston's brown mouse	hypothetical protein STEG23_014591 [Scotinomys teguina]	175	175	100%	9e-54	85.45%	145	KAL6031003.1
⊞1 member(s), 1 organism(s)	other sequences	SUMO-Lispro proinsulin fusion protein [synthetic construct]	176	176	78%	1e-53	97.67%	207	UIK24472.1
⊞3 member(s), 2 organism(s)	rabbits and hares	insulin precursor [Oryctolagus cuniculus]	173	173	100%	2e-53	86.36%	110	NP_001353985.1
⊞1 member(s), 1 organism(s)	coppery titi	Insulin [Plecturocebus cupreus]	175	175	100%	2e-53	87.27%	184	KAL0628984.1
⊞1 member(s), 1 organism(s)	E. coli	hypothetical protein [Escherichia coli]	172	172	95%	3e-53	84.76%	109	HDJ0545968.1
⊞1 member(s), 1 organism(s)	Chinese hamster	Insulin [Cricetulus griseus]	177	177	100%	3e-53	86.36%	248	EGW08477.1
⊞1 member(s), 1 organism(s)	Bank vole	hypothetical protein U0070_013151 [Myodes glareolus]	172	172	100%	6e-53	84.55%	135	KAK7801657.1
⊞2 member(s), 2 organism(s)	New World monkeys	insulin isoform X1 [Aotus nancymaae]	173	173	100%	6e-53	86.36%	155	XP_064226392.1
⊞2 member(s), 2 organism(s)	primates	insulin isoform X1 [Otolemur garnettii]	171	171	100%	8e-53	84.55%	110	XP_003798420.1
⊞4 member(s), 4 organism(s)	New World monkeys	RecName: Full=Insulin; Contains: RecName: Full=Insulin B cha…	171	171	100%	9e-53	85.45%	108	P67972.1
⊞8 member(s), 7 organism(s)	root	insulin-1 preproprotein [Rattus norvegicus]	171	171	100%	1e-52	82.73%	110	NP_062002.1
⊞1 member(s), 1 organism(s)	other sequences	NanoLuc-P2A-Insulin [Vector pJH2036]	177	177	100%	2e-52	81.82%	332	XOB88773.1
⊞1 member(s), 1 organism(s)	black flying fox	insulin [Pteropus alecto]	170	170	100%	3e-52	84.55%	110	XP_006911039.1
⊞10 member(s), 10 organism(s)	placentals	insulin precursor [Canis lupus familiaris]	170	170	100%	3e-52	88.18%	110	NP_001123565.1
⊞36 member(s), 29 organism(s)	root	insulin-2 preproprotein [Mus musculus]	169	169	100%	5e-52	81.82%	110	NP_001172012.1
⊞4 member(s), 4 organism(s)	primates	insulin [Eulemur rufifrons]	169	169	100%	5e-52	87.27%	109	XP_069327944.1
⊞1 member(s), 1 organism(s)	red fox	insulin isoform X2 [Vulpes vulpes]	169	169	100%	6e-52	88.18%	124	XP_072614830.1
⊞1 member(s), 1 organism(s)	North American porcupine	insulin [Erethizon dorsatum]	168	168	100%	2e-51	80.91%	110	KAM6160390.1
⊞1 member(s), 1 organism(s)	white-tufted-ear marmoset	insulin isoform X1 [Callithrix jacchus]	169	169	100%	2e-51	85.45%	148	XP_035119627.2
⊞1 member(s), 1 organism(s)	striped hyena	insulin [Hyaena hyaena]	167	167	100%	4e-51	83.64%	110	XP_039081000.1
⊞1 member(s), 1 organism(s)	grey whale	Insulin [Eschrichtius robustus]	169	169	100%	5e-51	88.18%	177	MBW01439.1
⊞1 member(s), 1 organism(s)	other sequences	proinsulin glargine analogue, partial [synthetic construct]	166	166	77%	8e-51	94.38%	105	AWO14628.1
⊞4 member(s), 4 organism(s)	rodents	insulin-1-like [Apodemus sylvaticus]	166	166	100%	1e-50	78.18%	110	XP_052028151.1
⊞1 member(s), 1 organism(s)	Iriomote cat	insulin isoform X1 [Prionailurus iriomotensis]	167	167	100%	2e-50	81.82%	171	GAB5578225.1

Figure 3: Top Homologous Sequence Clusters (Significant Alignments)

3.1 Summary of Top Homologous Hit Clusters

Organism / Species	Max Score	Query Cover	E-value	Per. Identity	Accession ID
Pan troglodytes (Chimpanzee)	221	100%	1e-72	98.18%	NP_00108996.1
Pan paniscus (Bonobo)	224	100%	2e-72	99.09%	XP_034787832.1
Macaca nemestrina (Macaque)	221	100%	4e-72	98.18%	XP_011719619.2
Chlorocebus sabaeus (Green monkey)	218	100%	5e-71	97.27%	XP_008002825.1
Pongo abelii (Orangutan)	191	100%	1e-60	88.18%	PNJ16423.1

4. Conserved Domain Architecture

Graphical domain mapping performed via the NCBI Conserved Domain Database (CDD) identified that the query sequence spans the ILGF_like (Insulin-like Growth Factor) superfamily architecture.

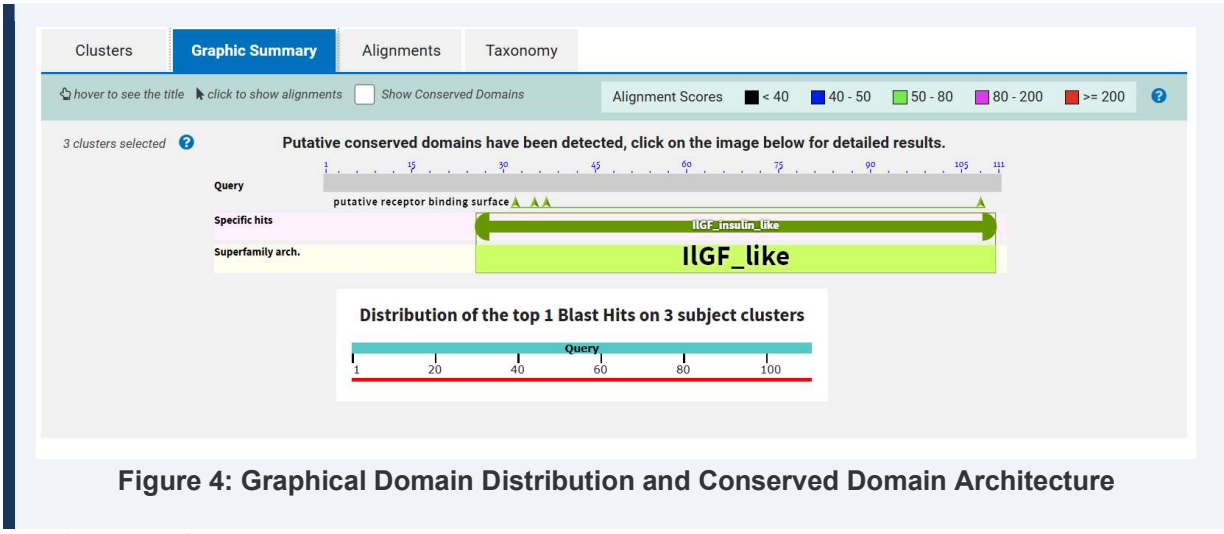


Figure 4: Graphical Domain Distribution and Conserved Domain Architecture

Key features of the domain mapping include:

- **Specific Hit Domain (ILGF_insulin_like):** Encompasses residues ~24 to ~108, matching the structural framework of the mature insulin peptide and C-peptide regions.
- **Putative Receptor Binding Surface:** Identified specific conserved contact residues (marked by green triangles in CDD display) critical for high-affinity binding to the Insulin Receptor (INSR).

5. Pairwise Sequence Alignment Interpretation

A detailed pairwise alignment was conducted between Human Insulin (Query) and Chimpanzee Insulin Preproprotein (Sbjct, Accession: NP_00108996.1).

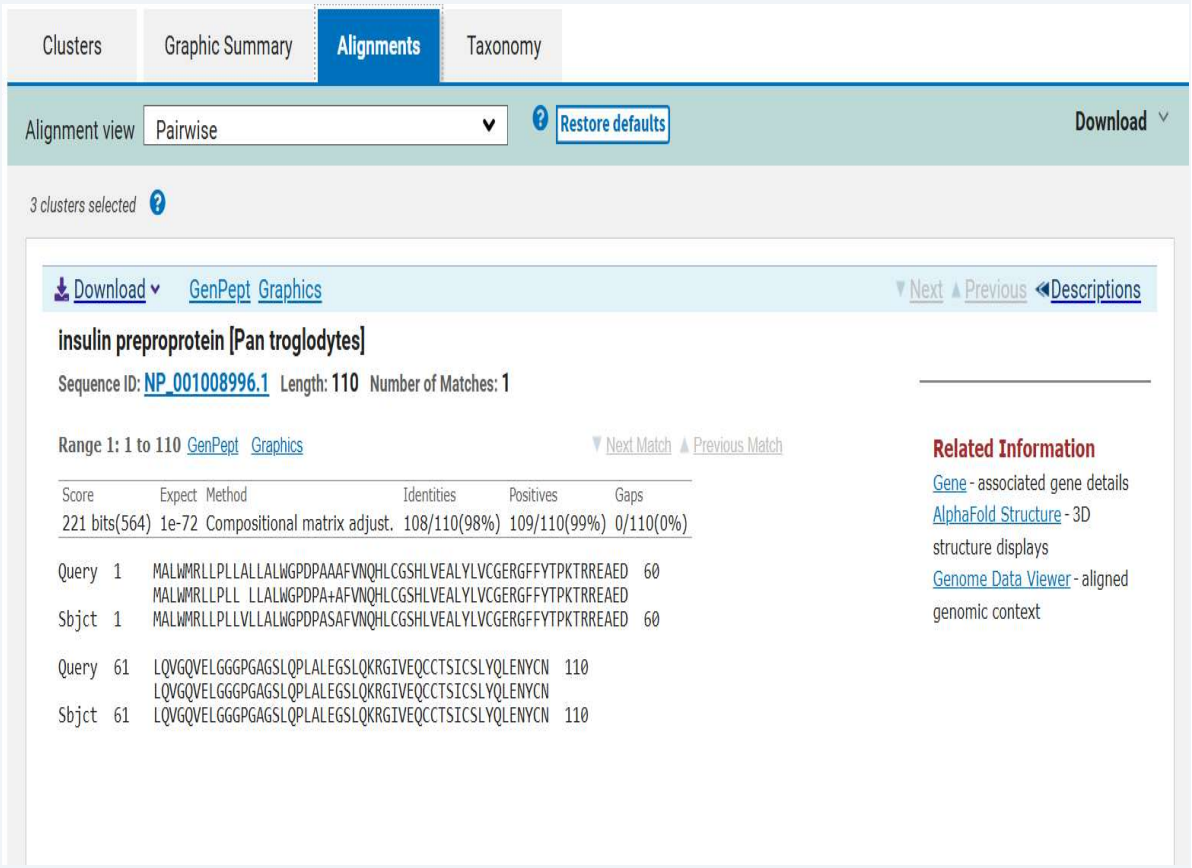


Figure 5: Pairwise Sequence Alignment (Homo sapiens vs. Pan troglodytes)

5.1 Pairwise Statistical Metrics

- **Bit Score:** 221 bits (564)
- **Expect Value (E-value):** 1e-72 (Statistical significance near absolute certainty)
- **Identity Percentage:** 108 / 110 (98.18% identity)
- **Positives (Physicochemical Similarity):** 109 / 110 (99.09% similarity)
- **Gaps:** 0 / 110 (0% gap insertion/deletion)

5.2 Sequence Alignment Text Output

```
Query  1   MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAED  60
           MALWMRLLPLLALLALWGPDPA AFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAED
Sbjct  1   MALWMRLLPLLVLLALWGPDPA SAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAED  60

Query  61  LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN  110
           LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN
Sbjct  61  LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSICSLYQLENYCN  110
```

Interpreting the alignment reveals that the sequence variation between Homo sapiens and Pan troglodytes is confined to minor conservative substitutions in the signal peptide (e.g., Alanine to Valine at position 15). The functional regions corresponding to the active hormone (Chain B residues 25-54 and Chain A residues 90-110) display 100% identity.

6. Discussion & Conclusion

The BLAST analysis of human preproinsulin (P01308.1) confirms severe evolutionary conservation across mammalian phylogenies. An E-value of 1e-72 along with 98.18% sequence identity with non-human primates underscores strong purifying selection acting on the insulin locus. The structural integrity of the ILGF_like superfamily domain is strictly preserved to ensure precise binding kinetics with the insulin receptor signaling apparatus.

References / Data Sources:

1. NCBI Basic Local Alignment Search Tool (BLAST): https://blast.ncbi.nlm.nih.gov/
2. UniProt Consortium. *UniProt: the Universal Protein Knowledgebase in 2023*. Nucleic Acids Res. 2023;51(D1):D523-D531.
3. Altschul, S. F., et al. *Basic local alignment search tool*. J. Mol. Biol. 1990;215(3):403-410.